Mad Scientist 1

Four calculated leads toward COP greater than one

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2026-10-09

Abstract

We computed preparation, loaded operation and reset for four candidate mechanisms: a falling-weight lever, switched transformer, contact chain and coil release. The lever delivered 1.1666 times its lifting input before reset, and 0.9951 times all input over the complete history. Transformer commutation reduced gross input by 6.90% at matched receiver delivery. Contact geometry and coil-release timing changed useful delivery substantially. The graphs and tables connect these gains to their actual transfers, giving the search for COP greater than one a quantitative starting point. The tested models' complete-history COP values remain below one.

PDF created: 2026-10-09 15:51:27 UTC
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Latest on GitHub: https://github.com/hobnilre/physics-mad-scientist-1

The top four

I want useful output, a repeatable machine state and a complete input record. The ambition is COP greater than one. Here COP means useful receiver work divided by **all gross inward energy**, including preparation, control and reset. The calculations below follow that ambition through four concrete mechanisms.

The list alternates the two leading mechanical and two leading electrical questions, preserving their order within each domain. This is an investigation priority, not a prediction of performance.

1. **The falling-weight lever.** Timed descent and compliance concentrate motion into a loaded takeoff arm. [OP-CT-HW-01]
2. **The switched transformer.** Physical commutation changes the receiver's return path and the source-work history. [OP-TRF-02]
3. **The contact chain.** Mass order and gaps reshape the passage of motion into a terminal load. [OP-LR54-03]
4. **The coupled-coil release.** Release phase, clamp voltage and winding orientation determine where a prepared electrical state goes. [OP-LR48-03]

We integrated 69 initial parameter cases and 17 exploratory coil extensions, then refined the leading cases and compared independent numerical methods. These are new finite model calculations, with exact examples explaining the observations that motivated them. Mechanical receivers absorb force–velocity work; electrical receivers collect resistor heat. All receiver energy stays outside the machine boundary. Runs began at rest with zero electrical and gate states.

The affirmative case for pursuing COP greater than one starts with resolved transfer gains: a loaded above-one operating-window ratio, a smaller supply requirement at equal service, and release choices that more than double the coil receiver's yield. The full histories identify exactly how far those gains carry.

1 The falling-weight lever

A teeterboard turns downward arrivals into an upward launch. Timing, leg compression and board flexibility affect the outgoing motion. The motivating question is whether that concentration can become useful work exceeding the input needed to prepare and repeat it.

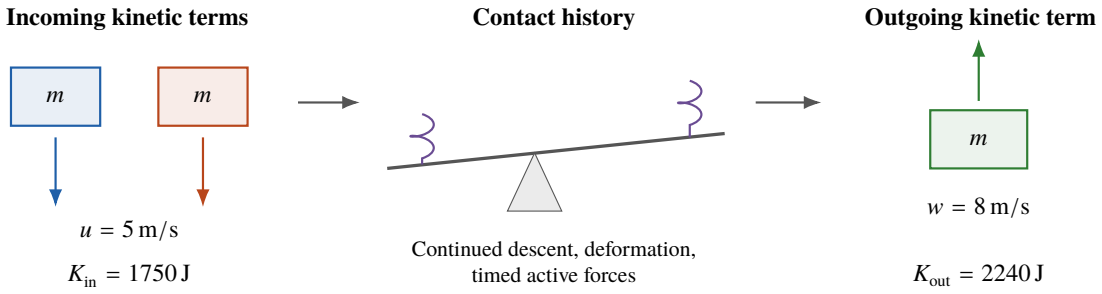


Figure 1. Illustrative equal-mass endpoint comparison. Continued descent and deformation are part of the contact history.

The observation. For two incoming masses and one outgoing mass, each 70 kg, speeds of 5 and 8 m/s give

$$K_{\text{in}} = 1750\,\text{J}, \qquad K_{\text{out}} = 2240\,\text{J}, \qquad K_{\text{out}}/K_{\text{in}} = 32/25. \tag{1}$$

The selected kinetic terms differ by 490 J. During contact, gravity continues doing work, springs compress, and active forces may act. The missing information is the trajectory connecting those endpoints. An ideal fixed pivot supplies impulse at zero work; it leaves the descent, rotation and contact forces to determine the delivery.

1.1 Calculated descent, delivery and return

We replaced the performers with two 1 kg vertical drivers at arm $a = -0.5$ m. The balanced rigid lever has inertia $I = 0.15\,\text{kg}\,\text{m}^2$, a 50 N m/rad torsion spring and bearing damping 0.1 N m s. Its opposite arm, $a_R = 0.8$ m, drives an outside viscous receiver: $v_R = a_R \cos\phi\,\dot{\phi}$ and $P_R = bv_R^2$. This loaded adaptation has its own scope alongside the human launch.

With upward driver heights y_i, contact compression is $\delta_i = a\sin\phi - y_i$ and $f_i = k[\delta_i]_+$, where $[z]_+ = \max(z, 0)$. The equations integrated were

$$\begin{aligned} m\ddot{y}_i &= -mg + f_i + u_i, \\ I\ddot{\phi} &= \tau_a - \sum_i f_i a\cos\phi - 50\phi - 0.1\dot{\phi} - b(a_R\cos\phi)^2\dot{\phi}. \end{aligned} \tag{2}$$

Actuator powers were $u_i\dot{y}_i$ and $\tau_a\dot{\phi}$. We used $g = 9.81\,\text{m/s}^2$.
The drivers rose 0.30 m during the first second, then released with a delay of 0, 0.10 or 0.20 s. Stiffnesses were 500 and 1500 N/m; receiver damping was 2, 8 or 32 N s/m. At 2.5 s, finite actuator trajectories returned the machine to its initial positions and velocities by 3.5 s. Preparation and return followed quintic position curves with continuous velocity; all their positive and negative work was integrated separately.

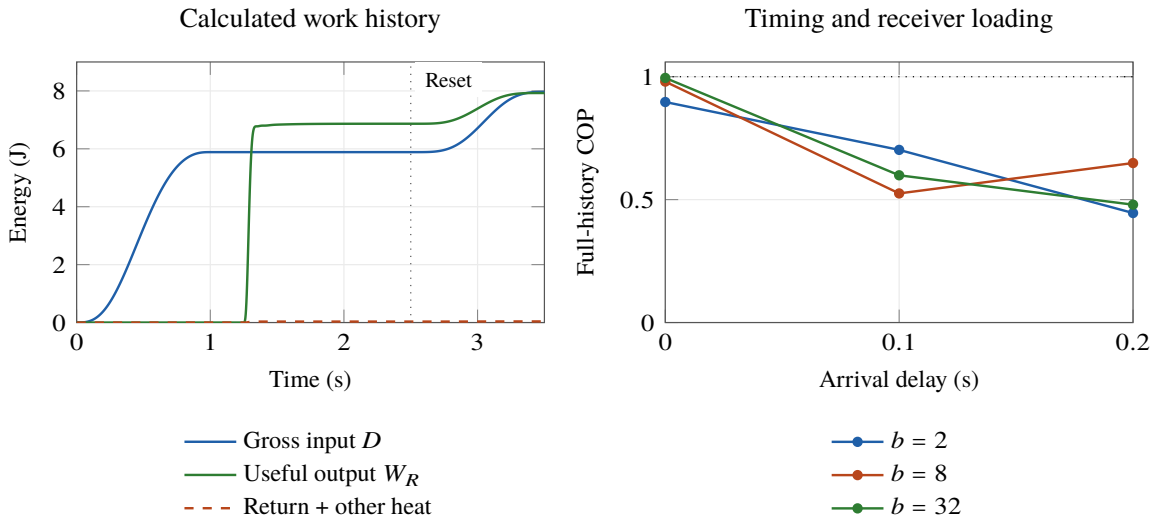


Figure 2. Computed lever results. Left: the best full-history case, including its driven reset. Right: the nine $k = 1500\,\mathrm{N/m}$ cases; b is in $\mathrm{N\,s/m}$. The dotted horizontal line is COP = 1.

What the calculation delivered. Simultaneous arrival with $k = 1500$ and $b = 32$ gave the highest COP among the 18 cases. Its lifting input was exactly $2mgH = 5.886\,\mathrm{J}$. Before reset, the receiver had collected 6.866664 J: a resolved operating-window ratio of **1.1666**. At that instant the machine store was −1.014425 J relative to its initial reference, including the drivers' lower gravitational positions.

Table 1. Computed work for the best lever case. Output during return is counted and identified explicitly.

Interval	Gross input (J)	Useful output (J)
Lift, 0–1 s	5.886000	0.000000
Descent/delivery, 1–2.5 s	0.000000	6.866664
Driven return, 2.5–3.5 s	2.080219	1.060545
Complete history	**7.966219**	**7.927209**

The full-history COP was **0.995103**. Other exported heat was 0.038999 J and actuator return 0.000011 J. With the same stiffness and simultaneous release, reducing b to 2 produced an upward arm speed of about 5.46 m/s but COP 0.8971. A 0.10 s delay at $b = 32$ reduced COP to 0.5990. The calculated lead is precise: contact timing gives strong motion and almost complete useful delivery in the best case, while restoration determines the remaining distance to above-one operation over the whole history.

2 The switched transformer

A tapped transformer offers different physical terminal pairs to a receiver. The motivating mechanical analogy is a gear train with several moving members: a receiver responds to the difference across its chosen connection. Electrical taps make the analogous voltage difference directly accessible.

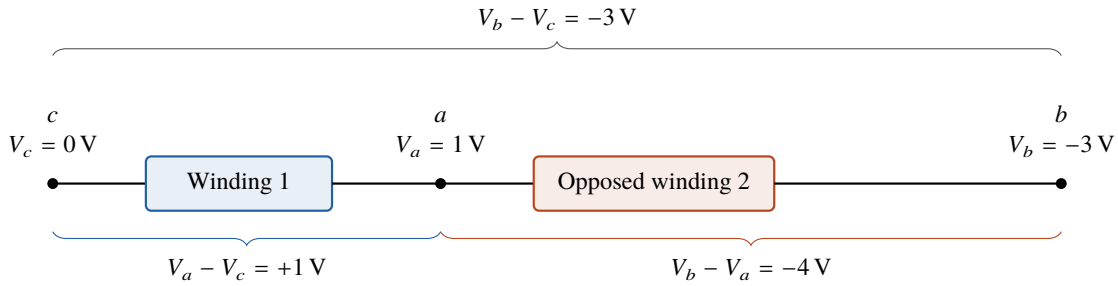


Figure 3. Ideal opposed-tap observation. The two terminal-pair voltages belong to one trajectory; a loaded return change creates a different circuit.

The observation. Winding 1 joins a, c and winding 2 joins b, a. For an opposed ideal turns ratio -4,

$$V_b - V_c = (V_b - V_a) + (V_a - V_c), \qquad V_b - V_a = -4(V_a - V_c). \tag{3}$$

At $V_a - V_c = 1\,\mathrm{V}$, the two readouts are -4 and $-3\,\mathrm{V}$. This invites a cycle that uses different connections at different times. A potential-zero shift leaves both differences unchanged; a physical return selector changes the currents and stored fields.

Fixed-tap descriptions leave that transition unspecified. Winding currents, stray charge, finite conductance and driver supplies participate in it. Earlier bounded switching comparisons do not give a universal supply ordering. We calculated an explicit finite selector to determine its ordering at a matched receiver service.

2.1 Calculated finite commutation

We grounded c and used nodes (a, b, r), each with 2 mF to ground. Currents run $a \to c$ and $b \to a$. The reciprocal winding matrix and series resistances were

$$\mathbf{L} = \begin{pmatrix} 0.05 & -0.19 \\ -0.19 & 0.8 \end{pmatrix}\mathrm{H}, \qquad \mathbf{R} = \mathrm{diag}(0.05, 0.2)\ \Omega, \qquad \mathbf{L}\dot{\mathbf{i}} + \mathbf{R}\mathbf{i} = \mathbf{A}^T\mathbf{v}. \tag{4}$$

The incidence matrix $\mathbf{A}$ has $+1$ at current entry and -1 at exit. Kirchhoff's node equations determine the three capacitor voltages. The source was $V_s = A_s \sin(4\pi t)$ through $0.2\,\Omega$ for $0 < t < 1$ s, then zero through the same resistor. The useful receiver resistor joins b, r.

The selector joins r to a or ground through $G = 0.001 + 49.999x$ siemens, where $\dot{x} = (\sigma - x)/(2\,\mathrm{ms})$. Each gate is a 1 μF capacitor driven through $2\,\mathrm{k}\Omega$ from 5σ volts; its supply power was included. During operation, $\sigma_a = 1$ when $\sin(4\pi t + \psi) \geq 0$ and $\sigma_c = 1 - \sigma_a$. Both commands went to zero at 1 s. A third gate added x siemens to each node's permanent 0.001 S bleeder from 1 to 10 s. Integration continued to 12 s.

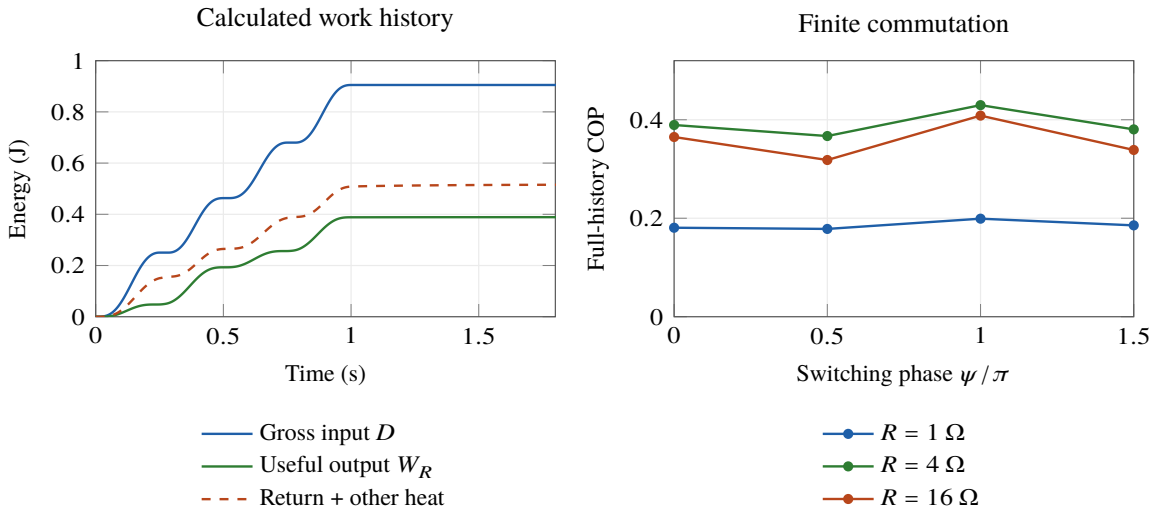


Figure 4. Computed transformer results. Left: switched $R = 4\,\Omega$, $\psi = \pi$; the early history is shown, while totals include the full 12 s. Right: all twelve commutated cases at unit source amplitude.

We computed fixed returns a, c and four phases $\psi \in \{0, \pi/2, \pi, 3\pi/2\}$ at receiver resistances 1, 4 and 16 Ω: 18 cases. Among switched cases at 4 Ω, phase π gave the highest COP. For the comparison below, we adjusted each fixed case's source amplitude and reran it to match the switched receiver's full-history 0.388864 J. Load and total interval remained the same; instantaneous receiver-power traces were allowed to differ. The fitted source amplitudes were 0.971127 V for fixed a and 1.106268 V for fixed c, against 1 V for commutation.

Table 2. Calculated matched-service comparison, including gate power and finite reset.

Return path	Gross input (J)	Useful output (J)	COP
Fixed a	0.972101	0.388864	0.400025
Fixed c	0.946703	0.388864	0.410756
Commutated, $\psi = \pi$	**0.905044**	**0.388864**	**0.429663**

Commutation reduced gross input by **6.90%** against fixed a, and **4.40%** against fixed c. Its source return was 0.004690 J and other heat 0.511491 J; final magnetic, capacitive and gate storage was below 10^{-20} J. Across all 18 unit-amplitude cases, fixed a at 16 Ω reached the largest COP, 0.433787.

This calculation supplies an affirmative result for the switching idea: physical commutation can improve useful delivery per unit of gross input. The gain belongs to a specified circuit and service comparison. Its complete work account also gives the size of the additional improvement required to reach the article's above-one target.

3 The contact chain

A disturbance passing through unequal masses can reverse one body and accelerate another. Contact release and re-engagement add a timing variable that a permanent spring does not possess. The COP question concerns how much of that rearranged motion reaches a useful receiver over a repeatable cycle.

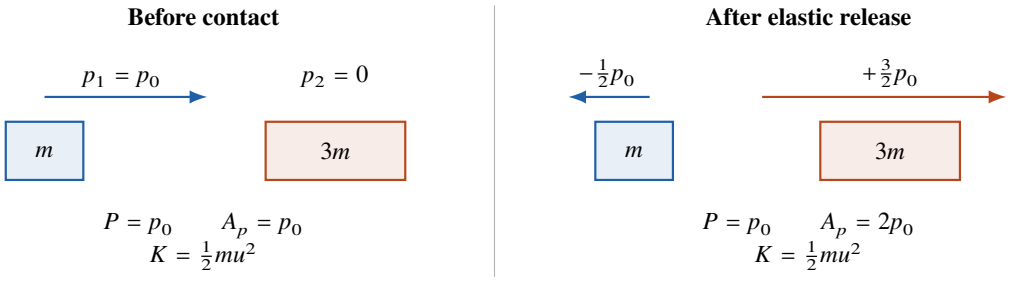


Figure 5. Exact elastic-contact example. Opposite momenta increase the magnitude sum while preserving the signed total.

The observation. A mass m arrives at speed $u > 0$ and strikes a stationary mass $3m$ through a lossless spring contact of stiffness k. During compression and recovery,

$$v_1 = \frac{u}{4}(1 + 3\cos\theta), \qquad v_2 = \frac{u}{4}(1 - \cos\theta), \qquad \theta = t\sqrt{\frac{4k}{3m}}. \tag{5}$$

At separation, $\theta = \pi$, the momenta are $-mu/2$ and $3mu/2$. Their magnitude sum has doubled; the signed sum is still mu. Endpoint kinetic energy is $mu^2/2$ in both states. We reproduced this exact trajectory numerically before adding more masses and a receiver.

The general identity $\sum_j |p_j| = \sum_j p_j + 2\sum_{p_j<0} |p_j|$ exposes the role of counterflow. For a free chain the signed sum stays constant, while active contacts change. Finite gaps, branching, restitution and fixture motion are the physical extensions behind the open question. Here the calculated extension is a loaded three-body line with gaps and a complete return trajectory.

3.1 Calculated contact and reset histories

The position deviations $\mathbf{x}$ start at zero. Each column of $\mathbf{B}$ has -1 at the left body and $+1$ at the right body of a contact. With gap offsets $\mathbf{d}$, the integrated equations were

$$\dot{\mathbf{x}} = \mathbf{v}, \qquad \mathbf{M}\dot{\mathbf{v}} = \mathbf{B}\mathbf{f} + \mathbf{u} - \mathbf{e}_3 b v_3, \qquad f_e = k[-\mathbf{B}^T\mathbf{x} - \mathbf{d}]_{e,+}. \tag{6}$$

The outside receiver absorbs $P_R = bv_3^2$; there is no internal damping in this model. Contact stiffness was 400 N/m. We computed mass sets $(1, 3, 1/2)$, $(1/2, 3, 1)$ and $(1, 1, 1)$ kg, gap pairs $(0, 0)$ and $(0.05, 0.10)$ m, and $b = 0.5, 2, 8$ N s/m: 18 cases.

The first body received $8\sin^2(\pi t/0.2)$ newtons for $0 < t < 0.2$ s. It then moved without launch drive until 2.5 s. Finite quintic actuator trajectories returned all bodies to their starting positions and zero velocity by 3.5 s. We integrated each actuator's work, including its returned work. Kinetic and unilateral spring stores were evaluated independently.

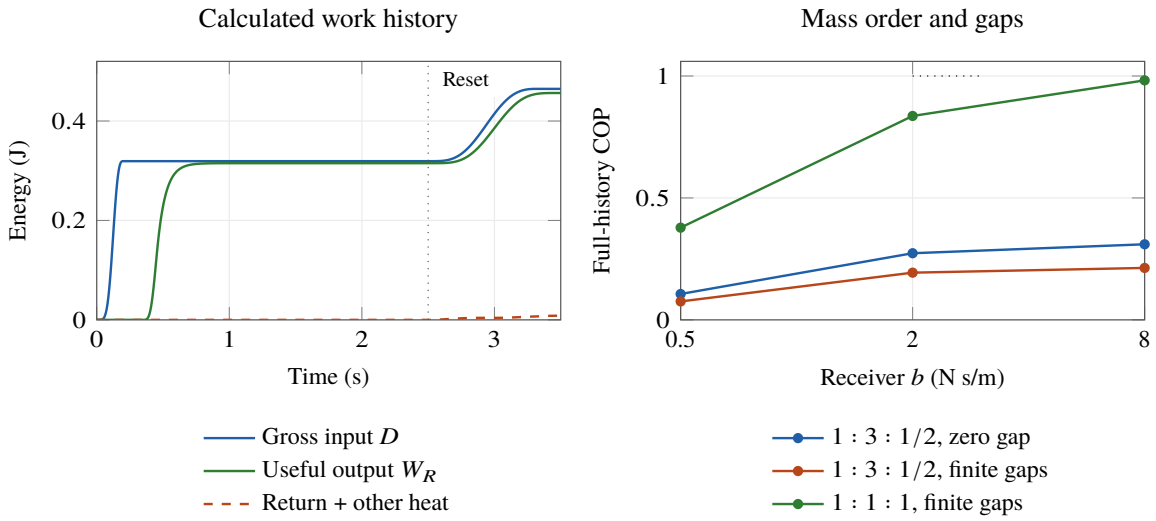


Figure 6. Computed chain results. Left: the best complete-history case, with equal masses and finite gaps. Right: unequal-mass and equal-mass controls under the same launcher and reset schedule.

What changed under load. The table compares three cases at $b = 8$. Here W_{pre} is all receiver delivery before reset, including any delivery during the launcher pulse. D_{reset} is gross reset input.

Table 3. Numerical comparison of loaded contact chains. A larger pre-reset output need not give a larger full-history COP.

Masses / gaps	W_{pre} (J)	D_{reset} (J)	Full COP
1 : 3 : 1/2, zero	0.162612	0.971015	0.309916
1 : 3 : 1/2, finite	0.239603	1.997999	0.212969
1 : 1 : 1, finite	**0.315183**	**0.145422**	**0.982149**

For the unequal chain, the finite gaps raised pre-reset delivery by about **47.3%**. The same force pulse supplied 0.196089 J without gaps and 0.319239 J with gaps, and the latter state cost more to restore. In the equal-mass control, launch input was 0.319246 J and total input 0.464668 J. Useful delivery was 0.456373 J, including 0.141190 J during reset; actuator return was 0.008295 J. All mechanical endpoint stores returned to numerical zero.

The useful result is the separation of three effects: momentum redistribution, loaded delivery and restoration work. The exact counterflow example remains a strong clue about transport, while the best computed cycle favored mass matching. Gaps improved one output window by nearly half. The complete trajectories show which part of that improvement contributes to the COP ambition and which part requires a better return path. Branches and dissipative contact laws remain outside this calculated line model.

4 The coupled-coil release

Two inductors exchanging through capacitance form an electrical counterpart to the contact chain. Their flux linkages can acquire opposite signs while the signed sum stays fixed. The idea is to change the circuit at a favorable phase and route the prepared state into a useful receiver.

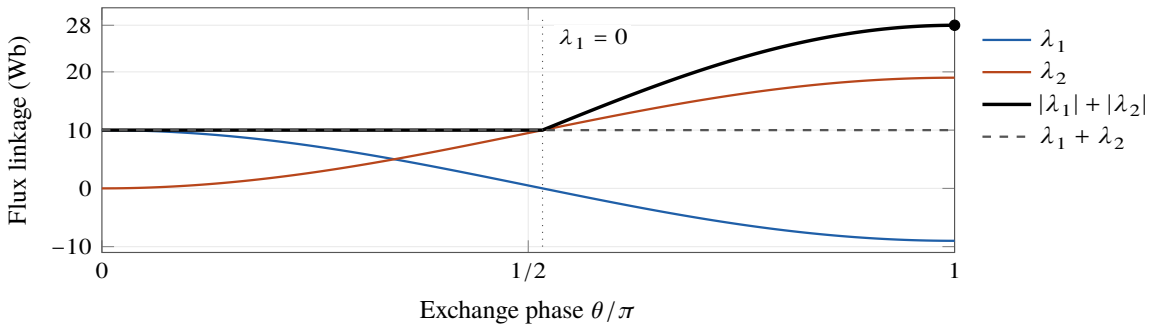


Figure 7. Exact connected-model reference. Opposing linkage signs separate the growing magnitude sum from the fixed signed sum.

The observation. With zero mutual coupling, $L_1 = 1\,\mathrm{H}$, $L_2 = 19\,\mathrm{H}$, initial currents $(10, 0)$ A and zero capacitor voltage,

$$i_1 = \frac{1 + 19\cos\theta}{2}\,\mathrm{A}, \qquad i_2 = \frac{1 - \cos\theta}{2}\,\mathrm{A}, \qquad \theta = t\sqrt{\frac{1/L_1 + 1/L_2}{C}}. \tag{7}$$

At $\theta = \pi$, flux linkages change from $(10, 0)$ to $(-9, 19)$ Wb. Their magnitude sum rises from 10 to 28 Wb. The capacitor returns to zero voltage and magnetic storage is 50 J at both endpoints. Independent numerical integrations reproduced the closed form.

Release opens a different question. Winding current must continue through specified capacitance, clamp and receiver paths. The connected solution identifies phases worth examining; the finite release circuit determines their useful yield.

4.1 Calculated release, clamp and receiver work

Our finite model used $L_1 = 0.1$ H, $L_2 = 1.9$ H, winding resistances $(0.02, 0.04)\ \Omega$, and node capacitances $(0.0001, 0.01)$ F. Winding 1 runs from a to ground; winding 2 runs from ground to b. The currents obey

$$\begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \dot{\mathbf{i}} = \begin{pmatrix} V_a - 0.02 i_1 \\ -V_b - 0.04 i_2 \end{pmatrix}, \qquad U_M = \tfrac{1}{2} L_1 i_1^2 + M i_1 i_2 + \tfrac{1}{2} L_2 i_2^2. \tag{8}$$

Kirchhoff's equations evolve both node voltages. A finite switch connects a, b; a separate switch connects b through receiver R to ground. The clamp at a carries $I_c = \mathrm{sgn}(V_a)[|V_a| - V_c]_+/(1\ \Omega)$.

Each gate obeys $\dot{x} = (\sigma - x)/(1\,\mathrm{ms})$, implemented by 1 μF and 1 kΩ at 5σ volts. Exchange and receiver-switch conductances are $10^{-6} + 100x$ siemens. The preparation branch connects $10\sin^2(\pi t/0.4)$ volts to a through $10^{-6} + 5x$ siemens for $0 < t < 0.4$ s. Preparation then switches off and exchange switches on. At release, exchange switches off and the receiver on.

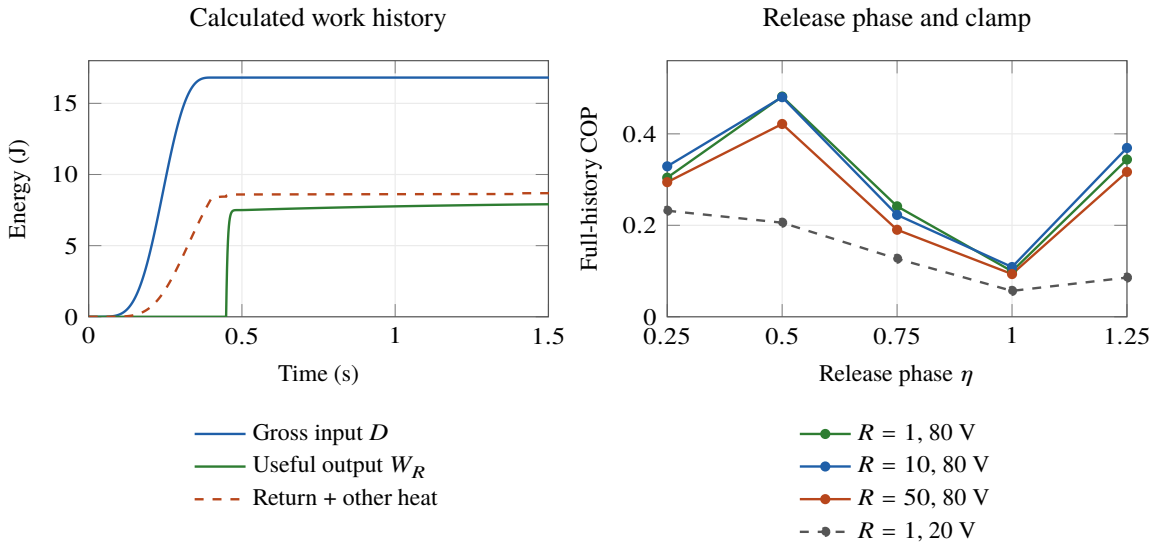


Figure 8. Computed coil results. Left: $M = 0$, 80 V clamp, $R = 1\ \Omega$, release phase $\eta = 1/2$; the early history is shown. Right: release/load comparisons. Every COP includes the full 14 s, gate supplies and reset.

Release occurred at $t_s = 0.4+\eta T_h$, where $T_h = \pi\sqrt{0.01/(1/L_1 + 1/L_2)} \simeq 0.0968304$ s. We computed five phases $\eta = 1/4, 1/2, 3/4, 1, 5/4$ and three loads $R = 1, 10, 50\ \Omega$. A $10^{-6}+0.1x$ S bleeder at each node was commanded on from 1.4 to 12 s. The receiver command also ended at 12 s; integration continued to 14 s.

The first 15 cases used a 20 V clamp, which conducted during exchange. A declared second 15-case exploration raised it to 80 V. Two further cases reversed signed coupling at the best zero-coupling phase and load.

Table 4. Computed coil histories at $R = 1\ \Omega$; $\kappa = M/\sqrt{L_1 L_2}$. All numbers are numerical approximations.

Clamp / η / coupling κ	Input (J)	Output (J)	COP
20 V / 1/4 / 0	16.8117	3.8995	0.2319
80 V / 1/2 / 0	**16.8117**	**8.0954**	**0.4815**
80 V / 1/2 / −1/2	19.6876	8.5788	0.4357
80 V / 1/2 / +1/2	19.6876	6.5646	0.3334

Changing clamp and release phase more than doubled the zero-coupling receiver work at essentially the same input. Waiting until $\eta = 1$ raised the linkage magnitude ratio to about 2.77, but gave COP 0.0995 at $R = 1$ and 80 V. The best energy delivery occurred earlier. Its other exported heat was 8.716310 J and its finite-reset storage remainder 5.0001 μJ. These results make the release path the productive design variable: the largest linkage-magnitude excursion and the best COP occur at different times.

What the calculations establish

The strongest calculated leads are a lever that crosses one in a loaded operating window, transformer commutation that saves input at equal receiver service, and coil timing that more than doubles useful delivery. They give specific mechanisms and scales to the ambition of COP greater than one. The whole-history account shows where each result stands.

Table 5. Best full-history COP in each explored family. These are selected model results, not universal optima.

Best tested model	Input D (J)	Useful W_R (J)	COP
Lever	7.966219	7.927209	0.995103
Transformer, fixed a	0.593317	0.257373	0.433787
Chain, equal masses	0.464668	0.456373	0.982149
Coil release	16.811682	8.095368	0.481532

One complete account. At every supply or actuator port, inward power P_k was integrated separately into gross draw $D_k = \int [P_k]_+ dt$ and returned work $X_k = \int [-P_k]_+ dt$. We used $D = \sum D_k$, $X = \sum X_k$, outside receiver delivery W_R and other exported heat H. Stored energy was evaluated from states, giving

$$r_E = \Delta U - D + X + W_R + H, \qquad \mathrm{COP} = \frac{W_R}{D} = 1 - \frac{X + H + \Delta U - r_E}{D}. \tag{9}$$

The receivers were passive, with no inward return. Shared supplies were counted once. Mechanical reset used work-metered forces and torques; electrical reset used the stated dissipative paths. Heat crossed an ideal bath boundary. Real controller losses and thermal dynamics would require additional laws.

What was checked. We integrated Newton/Kirchhoff equations rather than imposing energy closure. Refined leading cases used relative tolerance 2×10^{-10} and a maximum step of 3 ms. Independent explicit and implicit methods changed their COP by less than 3×10^{-10}. Their maximum sampled energy residuals were below 7×10^{-10} J. The coarser exploratory grid retained residuals up to 1.3×10^{-4} J.

Uniform power sampling initially missed a fast coil-switch transient by 0.178 J. Independent 8- and 16-point Gaussian integration within resolved solver steps reduced the leading cases' power-integral disagreements below 3×10^{-11} J. Exact contact/coil reference cases and symbolic component power identities supplied further checks. These are numerical sensitivity checks for the stated models, not certified error bounds or hardware uncertainty.

The case for continuing toward COP greater than one. Timing, topology and release laws changed useful output enough to justify treating them as design variables rather than fixed details. The computations also isolate the remaining question. For the passive laws used here, the derived power identities give $r_E = 0$; at a restored state, equation 9 leaves COP at or below one. Near-unity conversion by itself is not evidence of a crossing. An above-one complete-cycle argument therefore needs an independently supported physical contribution beyond these baselines. The calculated histories specify where to look for it: contact transfer and return, finite commutation, and the release/clamp path. That is the positive research case these results support.